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from google.colab import files
uploaded = files.upload()
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score
file_name = list(uploaded.keys())[0] # Automatically grab the uploaded file
df = pd.read_csv(file_name)

df = df.drop(['id', 'client_id', 'card_number', 'expires', 'acct_open_date'], axis=1)

df['credit_limit'] = df['credit_limit'].replace(['\$'], '', regex=True).astype(float)

label_cols = ['card_brand', 'card_type', 'has_chip', 'card_on_dark_web']
label_encoders = {}

for col in label_cols:
    le = LabelEncoder()
    df[col] = le.fit_transform(df[col])
    label_encoders[col] = le

X = df.drop('card_on_dark_web', axis=1)
y = df['card_on_dark_web']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("✅ Accuracy:", accuracy_score(y_test, y_pred))
print("\n📊 Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("\n📋 Classification Report:\n", classification_report(y_test, y_pred))

```



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Saving cards_data (2).csv to cards_data (2) (11).csv

✓ Accuracy: 1.0

Confusion Matrix:
[[1230]]

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	1230
accuracy			1.00	1230
macro avg	1.00	1.00	1.00	1230
weighted avg	1.00	1.00	1.00	1230

/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:407: UserWarning: A
warnings.warn(